MIL-R-21530/5D (WEPS)
26 February 1962
Superseding
MIL-R-21530/5C (WEPS)
29 August 1961

MILITARY SPECIFICATION
RESOLVER, ELECTRICAL, AC
FOR USE IN
POLARIS FLEET BALLISTIC MISSILE WEAPON SYSTEM
TYPE 11R99N4r8a

1. SCOPE

1.1 This specification establishes the requirements for a Resolver, Electrical, AC, Type 11R99N4r8a, hereinafter termed resolver.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-R-21530	Resolver, Electrical, AC, General Specification for

DRAWINGS

Bureau of Naval Weapons

1474170	Resolver, Type 11R99N4r8a

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 The resolver shall be in accordance with MIL-R-21530 with the following exceptions:

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(a) The resolver specified herein shall be tested as an RC only.

(b) The resolver specified herein shall not be tested for secondary impedance balance.

(c) The resolver shall not be tested for salt water corrosion.

3.2 The resolver shall be of the design, construction, and physical dimensions as specified on 1474170.

3.3 Specific characteristic functions.- The resolver shall have specific characteristic requirements as follows:

Input voltage	115 volts root mean square (v rms) ± 10 percent
Test voltage	115 v rms ± 2 percent
Test frequency	800 cycles per second (cps) ± 0.5 percent
Frequency range	400 and 800 cps
Primary	Stator
Temperature rise	10 degrees centigrade (C) max 15 gram centimeters at room temp 25 gram centimeters at -55 and +85 degrees C
Rotor impedance, Zro	See note
Stator impedance, Zso	See note
Stator impedance, Zss	See note
Total null voltage, maximum	50 millivolts (mv) at room temp 65 mv at -55 and +85 degrees C
Angular error versus voltage	20 seconds, max
Angular error versus frequency	20 seconds, max
Angular error versus vibration	90 seconds, max
Angular error versus humidity	45 seconds, max
Transformation ratio (Ke)	See note
Phase shift, stator to rotor	2.5 ± 1 degrees
Phase shift versus rotation	2.5 degrees max
Phase shift versus voltage	0.5 degrees max
Angular error, max	180 seconds at room temperature

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Null shift/para. 3.5.8 of MIL-R-21530.....	180 seconds at -55 and +85 degrees C
Angular interval.....	10 degrees
Impedance balance.....	60 Z-sec, max (resistive loads only)
Primary current, open circuit.....	0.010 amperes, nominal
Primary power, open circuit.....	0.121 watts, nominal
Sensitivity.....	1.28 volts per degree
Weight.....	4.7 ounces
Rotor moment of inertia.....	2 gram centimeters squared

Note: Requirements shall be as specified on Dwg. 1474170.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection shall be in accordance with MIL-R-21530.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-R-21530.

6. NOTES

6.1 The notes of MIL-R-21530 are applicable herein.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

PECIFICATION

MIL-R-21530/5D(WP) RESOLVER, ELECTRICAL, AC FOR USE IN POLARUS FLEET BALLISTIC

RGANIZATION *(Of submitter)* MISSILE WEAPON SYSTEM TYPE IIR9[illegible]8 AND STATE

ONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

ATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

IBMITTED BY *(Printed or typed name and activity)*	DATE

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DEPARTMENT OF THE NAVY
Bureau of Naval Weapons
Washington, D. C. 20360

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DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

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